The answer to this more refined question *is* yes. To avoid any confusion, let's spell this out. In Figure 6.1a, before Jupiter's gravity had any significant effect, the ball was heading purely to the right. Then, in Figure 6.1b, Jupiter's powerful gravitational force caught hold of the ball and pulled it toward the planet's center—a pull that's mostly downward but, as you can see in the figure, is also partially to the right. This means that as the ball closed in on Jupiter's surface, its rightward speed had increased somewhat, but its downward speed had increased dramatically. In the reverse run film, therefore, the ball's launch from Jupiter's surface would be headed somewhat *leftward* but predominantly *upward*, as in Figure 6.1c. With this starting velocity, Jupiter's gravity would have had its greatest impact on the ball's upward speed, causing it to go slower and slower, while also decreasing the ball's leftward speed, but less dramatically. And with the ball's upward speed rapidly diminishing, its motion would become dominated by its speed in the leftward direction, causing it to follow an upward-arcing trajectory toward the left. Near the end of this arc, gravity would have sapped all the upward motion as well as the additional rightward velocity Jupiter's gravity imparted to the ball on its way down, leaving the ball moving purely to the left with exactly the same speed it had on its initial approach.

All this can be made quantitative, but the point to notice is that this trajectory is exactly the reverse of the ball's original motion. Simply by reversing the ball's velocity, as in Figure 6.1c—by setting it off with the same speed but in the opposite direction—one can make it fully retrace its original trajectory, but in reverse. Bringing the film back into the discussion, we see that the upward-arcing trajectory to the left—the trajectory we just figured out with reasoning based on Newton's laws of motion—is exactly what we would see upon running the film in reverse. So the ball's time-reversed motion, as depicted in the reverse-run film, conforms to the laws of physics just as surely as its forward-time motion. The motion we'd see upon running the film in reverse is motion that could *really* happen in the real world.

Although there are a few subtleties I've relegated to the endnotes, this conclusion is general.[2] All the known and accepted laws relating to motion—from Newton's mechanics just discussed, to Maxwell's electromagnetic theory, to Einstein's special and general theories of relativity (remember, we are putting off quantum mechanics until the